

Ansible

Practical task

Use the documentation provided in the Learning Plan to complete the task above::

- a. Create ansible inventory with information about hosts
- b. Create ansible playbook for deploying spring-petclinic application. It should include:
 - i. Installing required software depending of application artifact (java based or docker based)
 - ii. Taking a specific artifact version (jar file/docker image) from a local machine or Nexus and deploying it to the application server.
- c. Deploy application to remote host with ansible
- d. Check results in browser

Deploy Spring Petclinic using Ansible:

Objective: Automate the deployment of the Spring Petclinic application using Ansible.

The playbook should:

- Create an inventory containing target hosts.
- Install the required software (Docker).
- Pull the Spring Petclinic Docker image from Nexus.
- Deploy the application on the remote server.
- Verify that the application is accessible through a web browser.

Prerequisites

Installed software:

- Docker Desktop
- Ansible
- SSH
- Nexus Repository Manager
- Spring Petclinic Docker image available in Nexus

Project Structure

```
spring-petclinic-ansible/
├── inventory
├── deploy.yml
├── ansible.cfg
├── vars.yml
└── vault.yml
```

Step 1: Create Inventory

Create an inventory file containing the remote server details.

```
[app_servers]
petclinic-server
```

```
[app_servers:vars]
ansible_host=localhost
ansible_port=2222
ansible_user=ansible
ansible_password=ansible
ansible_become_password=ansible
ansible_connection=ssh
ansible_python_interpreter=/usr/bin/python3
```

Since the remote host is an Ubuntu Docker container, SSH is exposed on port **2222**.

Verify connectivity: `ansible app_servers -i inventory -m ping`

```
rpavani@C19536 spring-petclinic-ansible % ansible app_servers -m ping
petclinic-server | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
```

Step 2: Create Playbook

Create `deploy.yml`.

The playbook performs:

- Install Docker
- Login to Nexus
- Pull Docker image

- Stop previous container (if exists)
- Run new Spring Petclinic container

```
---
- name: Deploy Spring Petclinic Application
  hosts: app_servers
  become: true

  vars_files:
    - vars.yml

  tasks:

    - name: Install Docker
      apt:
        name: docker.io
        state: present
        update_cache: yes

    - name: Login to Nexus Docker Registry
      shell: |
        echo "{{ nexus_password }}" | docker login {{ nexus_registry }} \
        -u "{{ nexus_username }}" \
        --password-stdin

    - name: Pull Spring Petclinic Image
      shell: |
        docker pull {{ docker_image }}

    - name: Stop existing container
      shell: |
        docker stop spring-petclinic || true
        docker rm spring-petclinic || true

    - name: Run Spring Petclinic Container
```

```

shell: |
  docker run -d \
    --name spring-petclinic \
    -p 8080:8080 \
    {{ docker_image }}

- name: Verify container is running
  shell: docker ps
  register: docker_output

- debug:
  var: docker_output.stdout_lines

```

Step 3: Execute Playbook

Run : ansible-playbook -i inventory deploy.yml

```

rpavani@C19536 spring-petclinic-ansible % ansible-playbook -i inventory deploy.yml
PLAY [Deploy Spring Petclinic Application] *****
TASK [Gathering Facts] *****
ok: [petclinic-server]
TASK [Install Docker] *****
ok: [petclinic-server]

```

Successful execution:

PLAY RECAP

petclinic-server

ok=8

changed=5

failed=0

Step 4: Verify Docker Image

SSH into a remote server.

```
ssh ansible@localhost -p 2222
```

Check images

```
docker images
```

Step 5: Verify Running Container

Run : docker ps

Step 6: Verify Application

Open <http://localhost:8080>

The Spring Petclinic homepage should appear.

